

Respiratory and Arousal Responses to Acoustic Stimulation*

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Study objectives: Although sleep-related obstructive apnea is most often associated with transient arousal, the impact of this arousal on respiratory control remains unclear. We tested the hypotheses that acoustic arousing stimulation can generate a significant respiratory response during sleep in healthy subjects and that the magnitude or timing of this response is affected by the presence of electrocortical arousal or inhaled carbon dioxide.

Design: We employed binaural tone bursts (0.5-s duration, 4-KHz center frequency, 99-s interstimulus interval) to elicit repetitive transient arousals from sleep during nocturnal polysomnographic recordings beginning at 10 PM and ending at 6 AM.

Participants: Recordings were conducted in five healthy adult volunteers aged 24 to 37 years.

Interventions: Inspired gas was alternated between room air and 3% to 7% CO₂ (titrated to yield an approximate 50% increase in minute ventilation) at 1-h intervals.

Measurements and results: Each 30-s epoch was scored for sleep/wake stage according to standard criteria. Only results obtained during nonrapid eye movement sleep are presented herein. Tone-evoked arousals were detected by computer analysis as increased EEG frequency occurring within 3 s of acoustic stimulation. For each tone, respiratory parameters for each of three prestimulus and four poststimulus breaths were normalized to the overall mean of prestimulus breaths measured during room air breathing for each subject. Tone bursts elicited repetitive transient arousals with a mean duration of approximately 10 s from all stages of sleep. With respect to the three prestimulus breaths, acoustic stimulation was associated with increased tidal volume and decreased inspiratory duration for at least four breaths. These respiratory responses to acoustic stimulation were not significantly influenced by either presence of transient arousal from sleep or inspired gas.

Conclusions: We conclude that transient EEG arousal may be repeatedly evoked from nonrapid eye movement sleep by transient acoustic stimulation in normal sleepers. This sensory stimulation is associated with augmented ventilation, a response that is not significantly affected by inspired hypercapnia or the presence of generalized EEG arousal. (CHEST 1997; 112:1567-71)

Key words: apnea; arousal; sleep

Abbreviations: ANOVA=analysis of variance; AR=EEG arousal; AS=acoustic stimulation; C1 to C3=three consecutive prestimulus control breaths before each acoustic stimulus; C₄/A₁=EEG lead derivation according to the international 10/20 system; HC=inspired hypercapnia; O₂/A₁=EEG lead derivation according to the international 10/20 system; R1 to R4=four consecutive poststimulus breaths after each acoustic stimulus; RA=room air; TE=expiratory duration; TI=inspiratory duration; VE=minute ventilation; VT=tidal volume

It has been observed that most sleep-related obstructive apneas resolve with accompanying EEG evidence of arousal (AR).¹⁻⁴ The relationships between AR and restored ventilation are unclear and probably quite complex, in part because the apnea and its resolution are accompanied by multiple

respiratory stimuli, including hypoxia, hypercapnia, and mechanoreceptor activation. The present study was designed to test the hypotheses that acoustic arousing stimulation can generate a significant respiratory response during sleep in healthy subjects and that the magnitude or timing of this response is affected by the presence of electrocortical AR or inhaled carbon dioxide.

MATERIALS AND METHODS

We studied five adults (four female, one male) between 24 and 37 years of age. None of the subjects was obese or had a history of cardiovascular, neurologic, or sleep-related disorders, and none was a habitual snorer.

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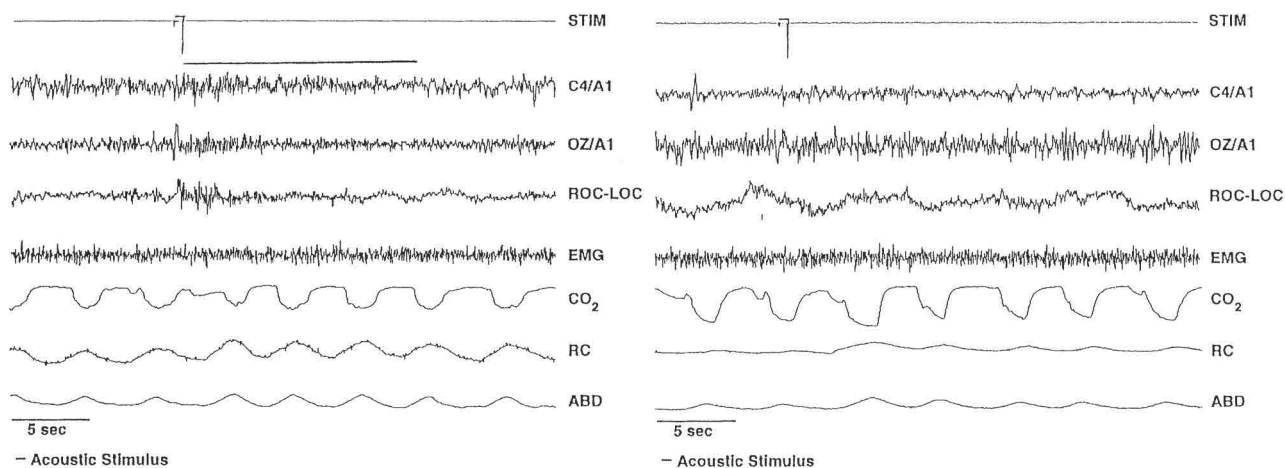


FIGURE 1. Two acoustic stimuli applied during nonslow-wave sleep during a single polygraphic recording. *Left*: EEG arousal following AS. Note the immediate EEG acceleration with alpha rhythm following the onset of the AS at the left. *Right*: AS without generalized EEG arousal. The dominant occipital EEG rhythm on the right is 3 to 4 Hz theta activity. Changes in the rate and depth of breathing follow both stimuli. The solid bar immediately above the C₄/A₁ EEG denotes the computer-detected arousal (see text for details).

Binaural tone bursts of 0.5-s duration, 4-KHz frequency, 85-dB SPL intensity, and 99-second interstimulus interval were administered using a tone generator (CA-1000; Nicolet Biomedical; Madison, Wis) and audiometer headphones (TD-100; Telephonics; St. Louis). Acoustic stimulation (AS) commenced after 5 min of continuous sleep and was suspended if the subject awakened for >90 s. AS commenced again after an additional 5 min of continuous sleep.

Central and occipital EEG leads, right and left referential electro-oculogram, and chin electromyogram were recorded by polygraph. Ventilation was monitored by respiratory inductive plethysmography calibrated by linear regression using an isovolume maneuver in the supine posture.⁵ Inspired and expired CO₂ at the nares, lead I ECG, arterial oxygen saturation, and an AS marker were also recorded on the polygraph. All signals were simultaneously digitized 100 times per second and stored on a personal computer (Experimenter's Workbench; Datawave Corp; Longmont, Colo) during the experiments. Inspired gas was alternated between room air (RA) and hypercapnia (HC) on an hourly basis throughout the recording, typically 10 PM to 6 AM. Hypercapnia was achieved by mixing 3 to 7% CO₂ in RA to achieve approximately 50% increase in inspired minute ventilation ($\dot{V}_E$). All inspired gases were administered via face mask at flow rates between 5 and 15 L/min which resulted in mask pressures of <2 cm H₂O.

Sleep was staged according to standard criteria.⁶ Figure 1 presents a typical response to AS. The solid bar (A) represents EEG arousal (AR) detected by computer as an increase in either central or occipital EEG centroid frequency of at least 2 SDs from the mean of the preceding 30-s epoch and with a duration of 3 to 15 s. Centroid frequency for each EEG derivation was computed at 1.28-s intervals using a 5.12-s window for power spectrum analysis. This definition of AR conforms to the American Sleep Disorders Association guidelines for EEG arousal,⁷ but has been implemented on a computer to improve objectivity and reproducibility. No AR occurred in Figure 1, *right*. Because an insufficient number of trials was obtained during nonrapid eye movement sleep in each subject (range, 0 to 14 trials), only data recorded during nonrapid eye movement sleep are presented herein.

For every tone, three control breaths (C1 to C3) preceding or during the tone and four response breaths (R1 to R4) following the tone were scored using the calibrated respiratory sum channel for measures of inspiratory time (T_I), expiratory time (T_E), tidal volume (V_T), and inspired $\dot{V}_E$. For each tone, the parameter value for each breath was divided by the grand mean of C1 to C3 for all tones administered during RA in that subject. If AR began within 3 s following the onset of the tone, it was scored as associated with the AS. Two groups of AS trials were accumulated for each inspired gas condition (RA, HC) in each subject; all stimuli that were followed by AR but not awakening or body position change were placed in the AR group, while all stimuli not followed by AR were placed in the non-AR group. For statistical assessment, analysis of variance (ANOVA) was performed with RA/HC, and AR/non-AR as factorial independent variables, with breath number (C1 to C3, R1 to R4) as an independent repeated measure and with T_I, T_E, V_T, and inspired $\dot{V}_E$ as dependent variables. Specific contrasts were tested and multiple comparisons were controlled by Fisher's protected least significant difference.

RESULTS

During RA breathing, 59 of 169 acoustic stimuli (35%) evoked arousals overall. This effect was de-

Table 1—Evoked EEG Arousals During RA and CO₂ Breathing in NREM Sleep*

	Latency, s	Duration, s
RA	0.88±0.35	10.4±0.7
CO ₂	0.95±0.29	11.64±0.7
P	0.88	0.49

*Values are mean±SE; p values are by paired *t* test. NREM=nonrapid eye movement.

Table 2—Effects of Hypercapnia on Baseline Respiratory Pattern*

	T _I , s	T _E , s	V _T , mL
RA	1.69±0.03	2.53±0.05	526±24
CO ₂	1.54±0.02	2.46±0.04	636±68
p	<0.0001	<0.0001	<0.0001

*Values are mean±SE for breaths C1, C2, and C3 pooled for all subjects. p values are by paired *t* test.

pendent on sleep stage: 42% in stages 1 and 2 vs 31% in stages 3 and 4 ($p<0.05$ by contingency table analysis). During hypercapnia, 60 of 136 tones (44%) elicited arousal ($p=0.05$ vs RA) with a sleep stage effect similar to that observed during RA breathing. Hypercapnia did not significantly affect either the latency from AS to the arousal or the duration of the arousal, as demonstrated in Table 1. As expected with respect to RA, hypercapnia was associated with shorter T_I and T_E as well as greater V_T ($p<0.0001$ by *t* test for each) resulting in the targeted increase in $\dot{V}_E$ (Table 2). Baseline respiratory pattern was equivalent prior to AS which did and did not evoke arousal ($p>0.05$ for all parameters by paired *t* test).

Figures 2 and 3 show plots of the pooled normal-

ized T_I and inspired $\dot{V}_E$, data, respectively. Four combinations of conditions are separately depicted for each parameter: RA with and without arousal, and CO₂ with and without arousal. Under no conditions were statistically significant changes in T_E observed. Changes in V_T were parallel to those illustrated for inspired $\dot{V}_E$. As depicted by Figure 2, during RA breathing, inspired $\dot{V}_E$ increased significantly on R1 ($p<0.0001$). The presence of arousal accelerated the return toward control values on RA (only R1 significantly different from control with arousal vs R1 and R2 different from control without arousal) and dampened the response during hypercapnia ($p=0.04$ for interaction between inspired gas and breath number). Figure 3 demonstrates the significant reduction in T_I that occurred on R2 ($p<0.0001$ for effect of breath number by ANOVA). The presence or absence of arousal did not affect the magnitude of decrease in T_I on R2 or the general pattern of T_I on subsequent response breaths ($p=0.77$ for effect of arousal). During hypercapnia, the T_I change on R2 and return toward baseline values followed a similar pattern ($p=0.92$ and 0.97 for interaction between inspired gas and breath number with and without arousal, respectively).

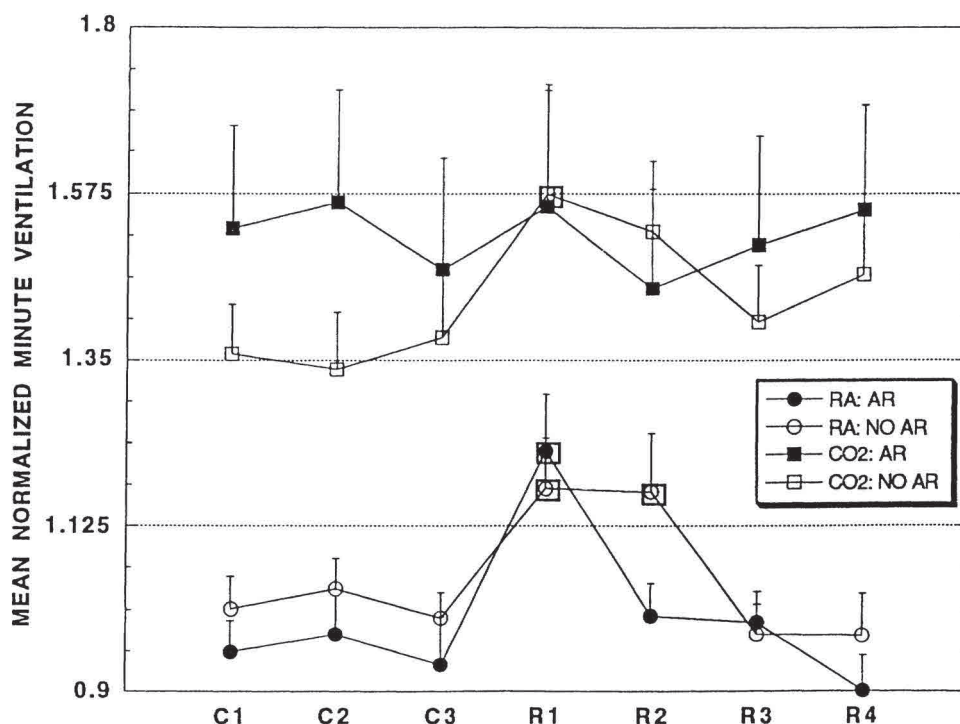


FIGURE 2. Average breath-by-breath changes in inspired $\dot{V}_E$ following AS. Data normalized to mean of all RA control breaths (C1 to C3) for each subject. Individual points represent mean±SE. Points enclosed by larger boxes indicate significant ($p<0.05$) differences with respect to control breaths. Significant increases in inspired $\dot{V}_E$ occurred on the first poststimulus breath (R1). The magnitude of this increase was independent of the presence or absence of AR (no significant interaction by ANOVA; $p>0.1$).

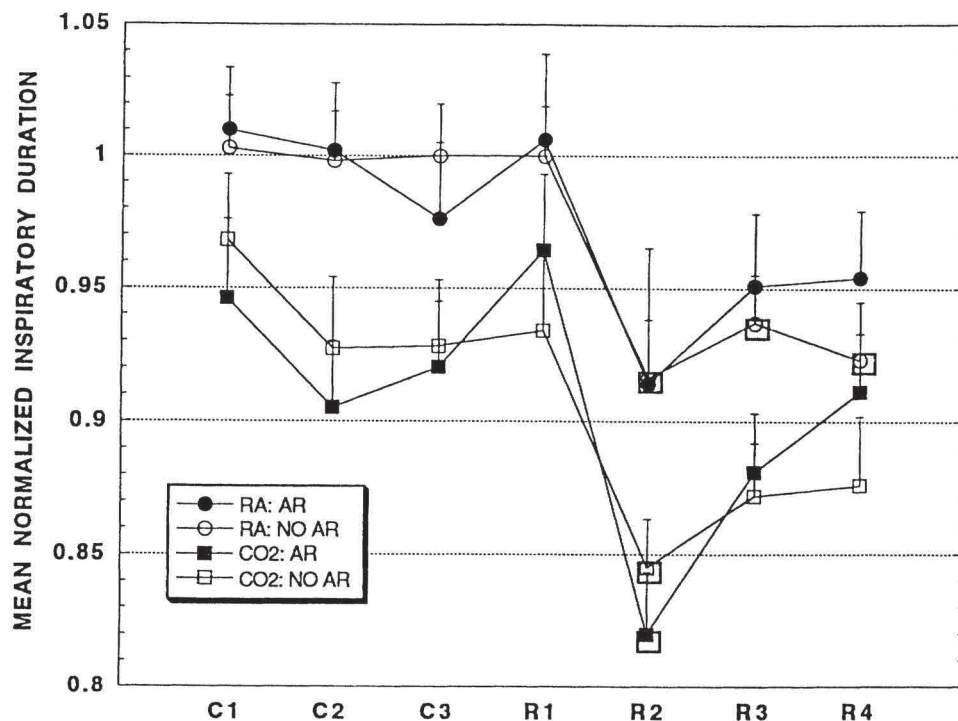


FIGURE 3. Average breath-by-breath changes in inspiratory duration following AS. Data are normalized as in Figure 2 (see text for details). Points enclosed by larger boxes indicate significant ($p < 0.05$) differences with respect to control breaths. A significant decrease in T_I occurred on the second poststimulus breath under all conditions, with no significant effect of inspired gas or arousal status on this response ($p > 0.1$ for each effect by ANOVA).

None of the effects described above differed significantly ($p > 0.05$ for all) between slow-wave (stages 3 and 4) and nonslow-wave (stages 1 and 2) sleep.

DISCUSSION

This study demonstrates that transient AS during normal nonrapid eye movement sleep evokes an immediate and significant increase in ventilation which, on average, lasts from one to four breaths following the stimulus. The acoustic is more often accompanied by generalized electrocortical arousal during stages 1 or 2 sleep than stages 3 or 4 sleep, but the respiratory response is largely unaffected by the presence or absence of transient EEG arousal.

It has long been speculated that ARs contribute to airway reopening in patients with obstructive sleep apnea syndrome.⁸ However, because of the simultaneous fluctuations in chemoreceptor and mechanoreceptor feedback that also attend airway occlusion, any independent contribution of arousal to airway reopening has never been clearly established. The present data clearly support the hypothesis that, in a background of respiratory equilibrium, isolated transient AS represents a significant respiratory stimulus. However, the presence of generalized electrocortical

arousal following the AS is not necessary for full elaboration of the respiratory response. Although the precise mechanisms underlying the respiratory responses cannot be determined from the present data, several inferences may be made.

Because the hypercapnic ventilatory response decreases during sleep with respect to wakefulness,⁹ we hypothesized that arousal may temporarily increase CO_2 -dependent respiratory drive toward awake values leading to a transient increase in ventilation. However, this hypothesis is not supported by the current observations. The fact that steady-state hypercapnia had no effect on the nature or magnitude of the respiratory response to AS suggests that this response is not mediated by a transient, state-dependent, increase in "effective chemosensitivity" of the respiratory controller. It seems most likely that in the context of the observed transient respiratory responses, the AS represented an independent drive that combined with the chemical stimuli in an additive fashion. The possibility that respiratory responses depend, at least in part, on state-dependent alteration in hypoxic respiratory drive cannot be ruled out. Arterial oxygen partial pressure or saturation was not measured in this protocol. However, because all subjects were healthy young adults, it is

unlikely that their baseline respiration was significantly dependent on hypoxic drive.¹⁰

Sequential responses to AS were observed in VT (breath R1) and TI (breaths R2 to R4). The immediate increase in VT is presumably a direct response to the AS. The subsequent decrease in TI may have resulted from the AS, the preceding increase in VT, or a combination of the two. According to the classical Hering-Breuer reflex, increased lung volume during inspiration tends to shorten the inspiratory phase, whereas lung inflation during the expiratory phase tends to lengthen TE. This reflex may provide at least a partial explanation for the previously described positive correlation between VT and TE of the same breath.¹¹ However, prolongation of TE was not observed following AS in the present protocol. The fact that TE for breath R1 and VT for breath R2 were both equivalent to control values argues against the possibility that increased activity of pulmonary or thoracic stretch receptors contributed to the decreased TI observed on breath R2 following AS. Similarly, if transient hypocapnia resulted from the augmented VE of breath R1, it would be expected to increase TI of subsequent breaths and thus offers no explanation for the observed decrease. Moreover, it is unclear that the Hering-Breuer reflex is active in man under normal physiologic conditions. Therefore, the decrease in TI appears to derive from the AS itself, rather than reflexively from the preceding augmented breath.

The finding that significant respiratory responses followed AS even in the absence of generalized electrocortical AR was unexpected and is of potential significance. We conclude that generalized electrocortical AR from sleep is not necessary for a significant respiratory response to be evoked by a nonrespiratory stimulus. In the light of this finding, it would be of interest to compare respiratory responses to AS during sleep and wakefulness. It may be that wakefulness would exert a damping, or blocking, effect on the respiratory response. AS was not performed during wakefulness in the present study.

Regional or localized cortical AR may have gone undetected in the present study. Only two EEG derivations were monitored. The montage is sufficient to identify global or generalized electrocortical ARs but must underrepresent the degree of cortical response following AS to some extent. It is also possible that specific cortical reflex arcs, which exerted descending influence on respiratory pattern generation, were activated by AS, which was below the threshold for generalized AR. Alternatively, the AS may have been coupled to respiratory control by

connections at the midbrain, pontine, or brainstem levels. The present data do not allow discrimination among these possibilities.

The present findings do suggest that it may be possible to significantly modulate respiratory drive via nonrespiratory afferent stimulation without causing significant sleep fragmentation or restriction. This may have potential therapeutic implications for disorders such as sleep-related hypoventilation and sleep apnea syndrome. In fact, we have recently demonstrated that AS with or without AR can shorten the duration of apnea in patients with obstructive sleep apnea syndrome.¹²

In summary, we have demonstrated that transient electrocortical state changes may be repeatedly evoked from nonrapid eye movement sleep using AS in healthy normal sleepers. This AS is associated with augmented ventilation, a response that is not significantly affected by HC or the presence of generalized EEG arousal.

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